$$\nabla \cdot \bar{E} = \rho/\epsilon \qquad \nabla \cdot \bar{B} = 0 \qquad \nabla \times \bar{E} = \frac{-dB}{dt} = -\mu \frac{dH}{dt}$$

$$\nabla \times \bar{H} = \bar{J} + \frac{d\bar{D}}{dt} = J + \epsilon \frac{dE}{dt} \qquad \bar{B} = \nabla \times \bar{A}$$

$$\nabla \times \bar{E} = -\frac{d\bar{B}}{dt} = -\frac{d}{dt}(\nabla \times \bar{A}) \Rightarrow \nabla \times \left(E + \frac{d\bar{A}}{dt}\right) = 0$$

$$\Rightarrow \bar{E} + \frac{d\bar{A}}{dt} = -\nabla V \rightarrow \text{electric scalar potential}$$

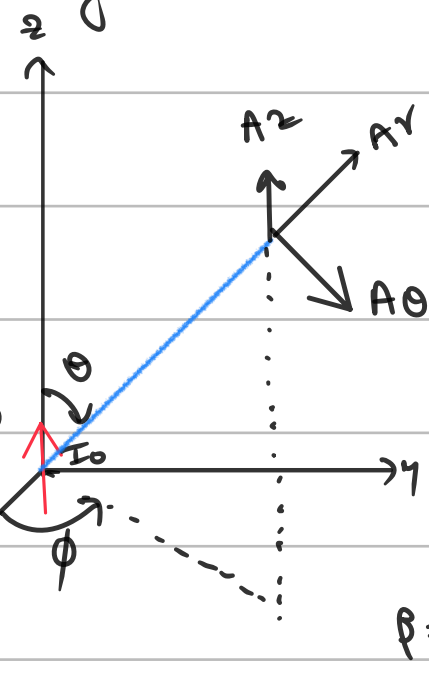
$$\nabla \cdot \bar{E} = \rho/\epsilon \Rightarrow \nabla\left(\frac{dA}{dt}\right) + \nabla^2 V = \frac{-\rho}{\epsilon}$$

$$\nabla(\nabla \cdot \bar{A}) - \nabla^2 A = \mu J + \mu\epsilon \frac{d}{dt}\left\{-\frac{d\bar{A}}{dt} - \nabla V\right\}$$

Source free $\Rightarrow \nabla \cdot \bar{A} = -\mu\epsilon \frac{dV}{dt}$ → Lorentz gauge condⁿ

$$\nabla^2 A = -\mu J \qquad \nabla^2 V = -\rho/\epsilon \rightarrow \text{poisson's eqⁿ}$$

Hertz dipole → infinitesimally small current element



$$I(t) = I_0 e^{j\omega t} \hat{z}$$

$$\bar{A} = A_z \hat{z} = \frac{\mu_0}{4\pi} \frac{I_0 e^{j\omega t}}{r} dL \cdot e^{-j\beta r} \hat{z}$$

$$A_r = A_z \cos\theta \qquad A_\theta = -A_z \sin\theta \qquad A_\phi = 0$$

$$H = \frac{1}{\mu}(\nabla \times A) \qquad H_r = 0 \qquad H_\theta = 0$$

$$H_\phi = \frac{I_0 dl e^{j\omega t} \sin\theta}{4\pi r} e^{-j\beta r}\left(j\beta + \frac{1}{r}\right) \qquad \beta = \omega\sqrt{\mu\epsilon}$$

$$E = \frac{1}{j\omega\epsilon}(\nabla \times H) \Rightarrow E_r = \frac{I_0 dl e^{j(\omega t - \beta r)} \cos\theta}{4\pi\omega\epsilon}\left(\frac{\beta}{r^2} - \frac{j}{r^3}\right)$$

$$E_\theta = \frac{I_0 dl\, e^{j(\omega t-\beta r)}}{4\pi\epsilon\omega}\sin\theta\left(\frac{j\beta^2}{r}+\frac{\beta}{r^2}-\frac{j}{r^3}\right)$$

$1/r \rightarrow$ radiation field $\quad 1/r^2 \rightarrow$ induction field
↳ far field
$1/r^3 \rightarrow$ electrostatic field $\rightarrow$ near field

$\rightarrow$ reactive fields

near field zone: $|E| = \frac{I_0 dl}{4\pi\omega\epsilon r^3}\sqrt{1+3\cos\theta}$

far field zone: $E_\theta = \frac{jI_0 dl\beta^2\sin\theta\, e^{j(\omega t-\beta r)}}{4\pi\epsilon\omega r}$

$E_\theta/H_\phi = \beta/\omega\epsilon = \sqrt{\mu/\epsilon} = \eta$ (intrinsic impedance of medium)

Power flow is solely because of radiation field

$$S_{avg} = \frac{1}{2}Re\{E\times H^*\} = \frac{1}{2}\left(\frac{I_0 dl\sin\theta}{4\pi r}\right)^2\frac{\beta^3}{\omega\epsilon}\hat{r}$$

$$P_{total} = \iint S_{avg}\cdot r^2\sin\theta\, d\theta\, d\phi = \frac{\pi}{3}\eta\left(\frac{I_0 dl}{\lambda}\right)^2 \qquad \eta=\sqrt{\frac{\mu}{\epsilon}}$$

Radiation Resistance: $\frac{2\pi}{3}\eta\left(\frac{dl}{\lambda}\right)^2$

Radiation pattern: plot of magnitude of radiation $\bar{E}$-field as function of θ and ϕ

$F(\theta,\phi) = \sin\theta$

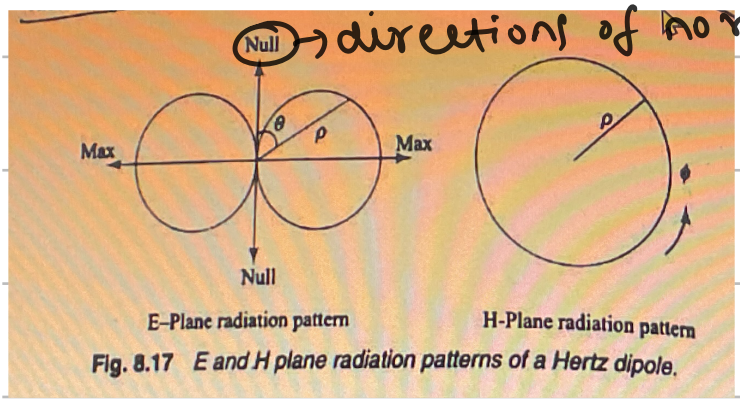
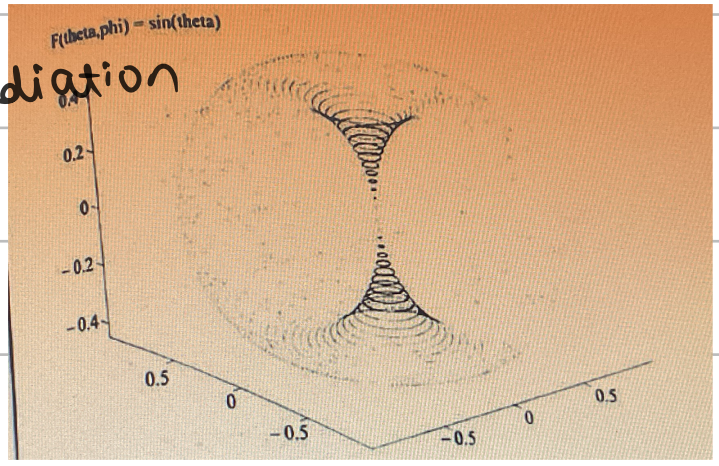


Fig. 8.17 E and H plane radiation patterns of a Hertz dipole.

→ directions of no radiation



Radiation parameters:

- main lobe → direction where radiation field strength is maximum
- Half power beam width: $\theta_1, \theta_2 \rightarrow E$ reaches $E_{max}/\sqrt{2} \Rightarrow HPBW = \theta_2 - \theta_1$. Pointing vector reduces by $1/2$ or $-3dB$
- Beam width btw first nulls: $\theta_{n_2} - \theta_{n_1}$ (nulls)
- directivity: quantifies radiation focusing capability of antenna

radiation intensity $U(\theta,\phi)$ = Power along (θ,ϕ) in solid angle $d\Omega$ / $d\Omega$

$= (\text{power density})\, r^2$

$$\frac{|E(\theta,\phi)|^2}{\eta} = S(\theta,\phi)$$

↓ pointing vector

Total power radiated = $\int U(\theta,\phi) \sin\theta \, d\theta \, d\phi$

Avg rad. intensity = $P/4\pi = \frac{1}{4\pi} \int U(\theta,\phi)\, d\Omega$

directive gain of antenna $G(\theta,\phi) = \frac{U(\theta,\phi)}{U_{av}} = \frac{4\pi U(\theta,\phi)}{\iint U(\theta,\phi)\, d\Omega}$

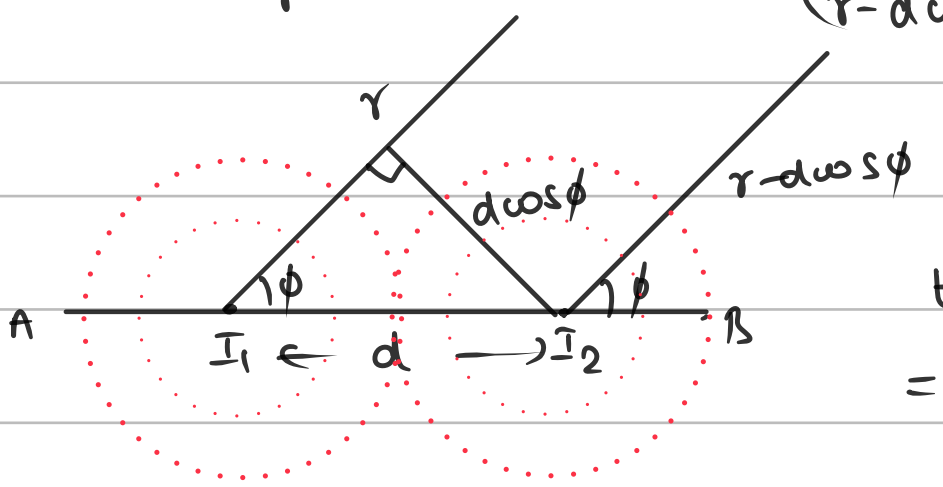
$D = \frac{U_{max}}{U_{avg}}$ (directivity) = $\max(G(\theta,\phi))$

$$= \frac{4\pi E_{max}^2}{\int_0^{\pi}\int_0^{2\pi} |E(\theta,\phi)|^2 \sin\theta \, d\theta \, d\phi} \qquad D \in [1, \infty)$$

Two element array

$I_1, \delta_1, I_2, \delta_2$ → two antennas currents & phases

$$E_1 = \frac{KI_1 e^{j\delta_1}}{r} e^{-j\beta r} \qquad E_2 = \frac{KI_2 e^{j\delta_2}}{(r - d\cos\phi)} e^{-j\beta(r - d\cos\phi)}$$



$$E = E_1 + E_2$$

$$= \frac{Ke^{-j\beta r}}{r} \left\{ I_1 e^{j\delta_1} + I_2 e^{j\delta_2} e^{j\beta d\cos\phi} \right\}$$

$$= K_0 I_1 e^{j\delta_1} \left\{ 1 + I_2/I_1 \, e^{j(\delta_2 - \delta_1)} \, e^{j \cdot \frac{2\pi}{\lambda} \cdot d\cos\phi} \right\}$$

parameters: d/λ, $\delta_2 - \delta_1$ ($= \delta$), I_2/I_1

- $I_2/I_1 = 1$ (given) $\Rightarrow \phi_{max} = \cos^{-1}\left(\frac{\delta\lambda}{2\pi d}\right)$
- $I_2/I_1 = 1$, $\delta_1 = \delta_2$ (given) $\Rightarrow \phi_{max} = \cos^{-1}(m\lambda/d)$

No of m allowed $= \lfloor d/\lambda \rfloor + 1$

Radiation pattern of non-isotropic elements =

Element radiation pattern x array factor (f(ϕ))